Capstone Proposal - Storefront Accessibility

Capstone Proposal: Monocular Measurement of Storefront Accessibility

Track: Technical Challenge (Track 2)
Team: [names — 2-4 people]
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Research Question

How accurately can entrance accessibility be measured from a single phone photo, without a depth sensor — and where does it break down?

The ADA specifies sharp numeric thresholds at building entrances: a vertical change up to 1/4” is acceptable, 1/4” to 1/2” must be beveled, anything over 1/2” requires a ramp, and an accessible doorway needs 32” of clear width. These are pass/fail lines measured in fractions of an inch.

That precision requirement is what makes this a real technical problem. A monocular depth model that is off by three inches produces an answer that is worse than useless, because it will confidently misclassify a compliant threshold as a barrier. A system accurate to a quarter inch would be genuinely useful. Nobody has published where the actual boundary sits.

We want to produce that answer: a calibrated error budget for single-image entrance measurement, plus a capture protocol that states the conditions under which the estimate can be trusted.

Why This Problem Is Hard

Monocular depth estimation recovers *relative* depth. Converting it to absolute inches requires either known camera intrinsics or a reference object of known size in frame, and error compounds through that conversion. Several things make entrances a worst case:

- **Oblique capture.** An untrained user photographs a door from wherever they are standing, not head-on. Angle error propagates non-linearly into height estimates.
- **Low-texture surfaces.** Concrete steps, painted metal thresholds, and glass doors give depth models very little to work with.
- **Scale ambiguity at small magnitudes.** The quantity we care about — a half-inch lip — is near the noise floor of the estimator, while the surrounding scene is measured in feet.
- **Glass and reflection.** Storefront doors are the exact material class that breaks depth estimation.
- **Occlusion and clutter.** Mats, sandwich boards, parked bikes, and pedestrians sit directly in the region of interest.

Prior Art and the Gap

We surveyed what exists before committing to this framing.

Crowdsourced accessibility maps — Wheelmap (~1M places rated directly, ~2M with partner data), AccessNow (200k+ places, 35 countries), WheeLog!, and Google Maps accessibility attributes. All of these are human-entered categorical judgments: a person looks at a door and picks green, yellow, or red. None of them measures anything, and coverage decays as businesses renovate.

Street-view detection research — Project Sidewalk and Tohme (UW Makeability Lab) established CV-based curb ramp detection with crowdsourced verification. Later work evaluated pavement condition, obstacles, and missing ramps across 300,000+ crowdsourced street images. ACAMAI (2025) trained a YOLOv8 detector on accessibility features and geolocated them across 360° Street View panoramas. RampNet appeared at an ICCV 2025 workshop on vision foundation models for accessibility. This line of work is **detection and classification at city scale** — “is a ramp present” — not metric measurement.

LiDAR commercial products — DeepWalk scans sidewalks and curb ramps with iPhone Pro LiDAR, builds 3D models, and automatically measures panel widths, slopes, and detectable warning surfaces for municipal ADA transition plans, with 120+ municipalities as customers. Apple’s Measure app does generic AR dimensioning but has no domain logic and is explicitly not a survey tool.

The gap. No one has characterized monocular metric measurement of building entrances. The existing work either (a) doesn't measure, (b) measures sidewalks rather than entrances, or (c) requires a LiDAR-equipped device and a trained operator following a deliberate capture protocol.

DeepWalk's existence is an asset rather than a threat to this project: **it validates that the measurement is worth having, and it gives us a ground-truth instrument.** Our question is how much of that capability survives the move to a single RGB frame on an arbitrary phone.

Assessment

Drawing the survey together, we assess the current state of the field as follows.

The need is established and unmet at the entrance. Accessibility information about buildings exists at scale, but it is categorical and human-generated. A wheelchair user consulting Wheelmap or AccessNow learns that someone once judged a place "partially accessible" — not whether the lip at the door is a quarter inch or two inches, which is the difference between entering and not. The measurement that determines the outcome is precisely the one no system captures.

Automated measurement is technically proven, but only under conditions that exclude most users. DeepWalk demonstrates that automated ADA measurement produces data municipalities will pay for. It does so with a depth sensor, on a restricted set of devices, operated by people trained to capture deliberately. That is a legitimate product decision and a real limit on reach: it cannot be run by an ordinary person standing on a sidewalk with the phone they own.

The research literature stops short of measurement. The Project Sidewalk line of work solved detection convincingly — locating curb ramps and obstacles across hundreds of thousands of street-view images. Detection answers a different question than measurement. Knowing a ramp is present says nothing about whether its slope complies, and no published work we found characterizes metric accuracy for entrance geometry from a single image.

We assess the open question as tractable but genuinely uncertain, which is why it is worth taking. Monocular depth models have improved substantially, and a reference object of known size collapses much of the scale ambiguity. Against that, the ADA thresholds sit at fractions of an inch, close to the noise floor of any monocular estimator, and realistic capture conditions are adversarial. We do not know whether the answer is yes or no. Establishing which — with error characterized by condition rather than reported as a single aggregate — is the contribution, and it holds either way.

We assess the project as feasible within the twelve-day window, on the condition that scope stays narrow. Data collection is unconstrained by access or permission, ground truth requires only a tape measure, and a LiDAR reference instrument is available. The binding constraint is time, not data, which is why the plan characterizes one measurement type completely rather than sampling several partially.

Proposed Solution

We propose to build and evaluate a **single-image entrance measurement pipeline** that takes an ordinary phone photograph of a storefront entrance and returns ADA-relevant dimensions with calibrated uncertainty.

The proposed system accepts one RGB image with no depth sensor requirement, identifies the entrance components within it, recovers absolute scale, and reports each measurement as an interval rather than a point estimate. Where the interval spans an ADA threshold, the system declines to classify rather than guessing. In proposing this, we are treating honest abstention as a design requirement: a confident wrong answer about whether a door is passable is worse for the user than no answer.

We propose to deliver this alongside the artifact that makes it credible — **a characterized error budget**. Rather than reporting one accuracy figure, the proposed evaluation stratifies error by capture angle, distance, lighting, surface material, and occlusion, so that the resulting capture protocol can state the conditions under which an estimate should be trusted. We propose to validate against two independent references: tape-measure ground truth on every entrance, and iPhone Pro LiDAR on a matched subset that establishes the accuracy ceiling a depth sensor would provide.

Finally, we propose a **direct comparison of two approaches to scale recovery** — anchoring on a reference object of known size in frame, versus deriving scale from camera intrinsics and estimated pose alone. These trade accuracy against usability in opposite directions, and quantifying that tradeoff is, in our assessment, the most transferable result the project can produce.

We are explicitly not proposing to ship a consumer accessibility map, replace crowdsourced platforms, or make compliance determinations of legal weight. The proposed contribution is a measured, honest answer to a bounded technical question, and the evidence that supports it.

Approach

The technical implementation of the proposed solution, in three stages.

Stage 1 — Region-of-interest detection. Segment the entrance into its components: door plane, threshold, step faces, ramp surface, and ground plane. Fine-tuned segmentation over a general vision backbone.

Stage 2 — Metric recovery. Two arms, compared head to head: - *Reference-object arm:* a known-size object in frame (credit card, standard brick course, printed marker) anchors absolute scale. - *Intrinsics-only arm:* EXIF focal length plus estimated ground plane and camera pose, no reference object.

Stage 3 — Compliance reasoning. Map measurements to ADA thresholds and emit a decision with a confidence interval, including explicit abstention when the estimate straddles a threshold. Abstention is a first-class output, not a failure.

The head-to-head between the two arms is the core ablation. The reference-object arm should be more accurate; the intrinsics-only arm is more usable. Quantifying that tradeoff is a real contribution.

Data and Ground Truth

The capture constraint drove the topic choice: **storefronts are publicly observable and require no permission or private access.**

- **Reference measurements:** tape measure on every entrance, recorded to 1/8”.
- **LiDAR reference:** iPhone Pro scans of a subset, to compare monocular error against depth-sensor error on identical scenes.
- **Test images:** shot the way an untrained person would — handheld, arbitrary angle, whatever the ambient light is. Deliberately *not* a clean capture protocol, since realistic capture is the condition we are evaluating under.
- **Target:** 100-150 entrances, each captured from 3-4 angles and 2 distances, across a range of materials (concrete, tile, metal threshold, glass door) and lighting.

Each image is tagged with capture conditions (angle, distance, lighting, material, occlusion) so error can be attributed to specific conditions rather than reported as one aggregate number.

Evaluation

Primary metrics: - **Absolute error in inches**, per measurement type (threshold rise, step height, clear width, ramp slope). - **Threshold classification accuracy** against the ADA pass/fail lines — the metric that actually matters. - **Error stratified by capture condition** — the deliverable most likely to be useful to others. - **Calibration** — do the confidence intervals contain the true value at the stated rate? - **Monocular vs. LiDAR** on the same scenes, establishing the ceiling.

Negative results are a valid outcome. If single-image estimation cannot reach the half-inch line, documenting exactly why and under what conditions it fails is a legitimate finding and we will present it as one.

Deliverables

1. Labeled dataset of storefront entrances with tape-measure ground truth and condition tags.
2. Working measurement pipeline, demonstrable live on a phone.
3. Error budget: accuracy by measurement type and capture condition.
4. Ablation: reference-object vs. intrinsics-only scale recovery.
5. Comparison against LiDAR on matched scenes.
6. Written findings and a capture protocol stating when the estimate is trustworthy.

Timeline (12 days)

Days	Work
Aug 28-30	Scope lock. Build capture rig and labeling schema. Collect first 30 entrances with tape ground truth. Baseline off-the-shelf depth model to establish the floor.
Aug 31-Sep 2	Complete dataset to target size. Stand up segmentation for entrance components. Both scale-recovery arms working end to end.
Sep 3-5	Evaluation harness. Full error analysis stratified by condition. LiDAR comparison subset.
Sep 6-7	Iterate on the dominant failure mode identified in evaluation. Freeze results.
Sep 8	Cursor Workshop. Build the live demo. Slide deck.
Sep 9	**Demo Day** — 10 min presentation, 5 min Q&A.

Twelve days is tight, so scope is deliberately narrow: **one measurement type is fully characterized before any second type is attempted.** Threshold rise is the priority, since it is the most common barrier and the hardest measurement. Everything else is a stretch goal.

Demo Day Presentation Plan

Mapped to the Technical Challenge requirements:

- **Research question** — can a phone photo measure an entrance to ADA precision?
- **Approach** — two-arm scale recovery, evaluated against tape and LiDAR.
- **Technical demo** — live measurement of an entrance in the room, plus the error analysis walkthrough.
- **What we learned / where next** — the error budget, the conditions that break it, and what would need to change to close the gap.
- **Slide deck.**

Live measurement at the venue is the demo we want: walk to a door in the building, measure it on stage, and compare against a tape measure in real time. It is a risk if accuracy is poor, but showing an honest confidence interval and an abstention is a stronger result than a rehearsed video.

Build-in-Public Plan (X)

Per-person targets by Sep 9: 150+ engagements, 25+ new followers, 5+ posts, 1+ outside repost.

This project has natural visual content, which is the main lever. Planned posts:

1. **Day 1** — the problem framed in one image: a half-inch lip that makes a store unusable for a wheelchair user, next to the ADA line that says it's a violation.
2. **Dataset build** — photo grid of entrances collected, tape measure in frame.
3. **First results** — the honest early error number, including if it's bad. Visible failure cases perform well.
4. **The ablation** — reference object vs. intrinsics, side-by-side error chart.
5. **LiDAR vs. single photo** — the money chart: how much accuracy you lose without a depth sensor.
6. **Demo Day** — live measurement clip.

For the outside-repost target, the accessibility and civic-tech communities on X are active and receptive, and the disability advocacy audience in particular engages with concrete measurement work. Tagging the Project Sidewalk / Makeability Lab research community is a reasonable ask given we are building directly on their line of work.

Open Questions for Our Mentor

1. Is the negative-result framing acceptable as a primary outcome, or should we hedge toward a guaranteed-positive deliverable?
2. Is 100-150 entrances the right dataset size for the time available, or should we trade breadth for more angles per entrance?
3. Should the intrinsics-only arm be cut if it looks weak by Sept 2, to concentrate effort on characterizing the

reference-object arm properly?

Risks

Risk	Mitigation
Monocular accuracy is far off the ADA thresholds	Reframe as a characterization result; the error budget is the contribution either way
Data collection eats the timeline	Dataset target is a floor, not a goal; 60 well-labeled entrances beats 150 sloppy ones
Segmentation on glass doors fails outright	Scope to opaque-door entrances first; treat glass as a documented failure class
Live demo fails on stage	Pre-recorded backup of the same measurement, shown only if needed